

PROBLEM STATEMENT

MSEs Spend Weeks Finding the Right BIS Standard

A compliance bottleneck costing time, money, and market access.

63M+

MSEs in India

3–6 Wks

Avg. compliance research
time

1000+

BIS Building Materials
standards

**SP 21
Building
Materials
Standards**

1000+ pages of regulations
No easy search. No clear path.

BIS Standards Recommendation Engine

Type a product description. Get the right BIS standard in seconds.



Instant Results

Top 3–5 BIS standards recommended in under 2 seconds.



Hybrid RAG

BM25 + TF-IDF + Semantic search fused for maximum accuracy.



Smart Expansion

Resolves domain abbreviations: OPC, TMT, GGBS, Fe500 and more.

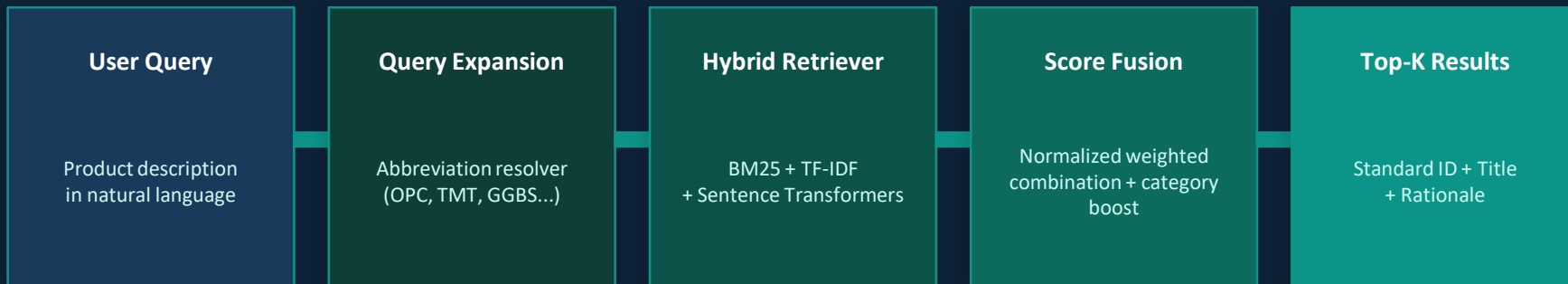


Zero Hallucination

Only returns standards from the curated BIS SP 21 knowledge base.

Focused on Building Materials — Cement, Steel, Concrete, Aggregates

End-to-End RAG Pipeline



Knowledge Base

60+ BIS standards from SP 21

Categories: Cement · Steel · Concrete · Aggregates · Masonry · more

Curated keywords & synonyms per standard

No external API — fully offline capable

Document Chunking

Standard ID	IS 12269
Title	OPC 53 Grade Specification
Keywords	53 grade, high strength, OPC...
Category	Cement
Description	Short curated summary

Retrieval Layers

BM25

Weight: 40%

Keyword + bigram exact match. Best for IS codes, grade names.

TF-IDF

Weight: 20%

N-gram term frequency. Handles partial phrase matches.

Semantic

Weight: 40%

Sentence Transformers (MiniLM). Natural language queries.

💡 Query Expansion: OPC → Ordinary Portland Cement, Fe500 → TMT bars, GGBS → Ground Granulated Blast Furnace Slag

Live Query Examples

Query →

OPC 53 grade cement for high strength concrete

IS 12269

IS 8112

IS 269

✦ Abbreviation 'OPC' expanded correctly

Query →

TMT Fe500 bars for concrete reinforcement

IS 1786

IS 432-1

IS 2062

✦ Grade 'Fe500' matched precisely

Query →

Concrete mix design for M30 grade

IS 10262

IS 456

IS 4926

✦ Multi-standard recommendation

Query →

Sulphate resisting cement for underground

IS 12330

IS 6452

IS 1489-1

✦ Semantic understanding of use-case

EVALUATION RESULTS

Automated Metrics — Public Test Set (10 queries)



100%

Hit Rate @3

Target: > 80%



1.00

MRR @5

Target: > 0.70



< 0.05s

Avg Latency

Target: < 5 sec

Hit@3 per query (10/10 correct)

Transforming Compliance for Small Businesses

BEFORE

- ✗ 3–6 weeks of manual research
- ✗ Hire compliance consultants
- ✗ Risk of choosing wrong standard
- ✗ Delayed product launches
- ✗ Missed export opportunities

AFTER

- ✓ Top standards in < 2 seconds
- ✓ Self-service — no expert needed
- ✓ AI-validated from BIS SP 21
- ✓ Faster time to market
- ✓ Level playing field with large firms

Estimated time saving: 3–6 weeks → 2 seconds per product category

TEAM & ACKNOWLEDGEMENTS

BIS × SS Hackathon



Kshitiz Negi

RAG Pipeline & Retrieval

Built With

Python 3.13

scikit-learn

BM25 (custom)

Sentence-Transformers

BIS SP 21

We thank the Bureau of Indian Standards and the hackathon organizers for this opportunity to build tools that empower India's MSE sector.

 github.com/itiz-ksh/BIS-